

AKSHAY KHULE

Hands-on Principal Architect | Java/Python Full Stack · Distributed Systems · Data Platforms · Generative AI

Mobile: 9326768815 | Email: khule.akshay@gmail.com

[Publications](#) | [GitHub](#) | [Cloud Certifications](#) | [Distributed Systems Certifications](#)



PROFESSIONAL SUMMARY

Hands-on Principal Architect | Python & Java backend engineering, distributed systems, and production GenAI.

- Global technical leadership across Software Engineering, Product Development, and Systems Architecture — architecting and shipping production-grade distributed systems, cloud-native microservices, petabyte-scale data platforms, and GenAI solutions across Banking, Pharma, AI/ops, Manufacturing, and Cybersecurity domains, owning end-to-end High-Level and Low-Level Design for scalable, resilient systems.
- Hands-on design and development of RESTful APIs, microservices, and serverless backends in Python (FastAPI, Pydantic, asyncio, SQLAlchemy, pytest) and Java (Spring Boot, Spring Cloud, Spring Security) , OAuth2/JWT with KeyCloak, RDBMS schema design, Elasticsearch/OpenSearch, Redis, and message queues (Kafka, SQS) — engineered with service discovery, centralized configuration, resiliency patterns, and distributed tracing; containerized with Docker and deployed on Kubernetes across AWS and Azure.
- Expert in end-to-end Big Data, Data Lake, and Data Platform architecture - streaming/batch ingestion, extraction, cleaning, transformation, enrichment, governance using PySpark (Structured Streaming, Pandas UDFs), Databricks (Delta Lake, Unity Catalog), Kafka, Azure Synapse, Azure Data Factory, Synapse Data Warehouse and Microsoft Fabric; automated deployment and data-quality validation with Python, delivering petabyte-scale, GDPR-compliant pipelines with zero-data-loss guarantees.
- Generative AI: Architected the multi-agent foundation for [Unisys Intelligence Accelerator](#) (UIA) — FastAPI-based agent runtime services exposing agent capabilities via REST, with a custom Python agent harness running on Amazon Bedrock AgentCore and Microsoft Foundry, grounded via managed Knowledge Bases (RAG) with embeddings and vector search — automating IT operations, app development, and data tasks for large enterprises.
- Contributed to technical strategy, architecture guardrails, and engineering standards adopted across multiple product teams (led teams of 30+); hired and mentored architects and engineers; delivered winning RFP responses for new business.

PUBLICATIONS & THOUGHT LEADERSHIP

- **White paper:** GenAI-Augmented Polyglot Persistence for Microservices - Toward Self-Adaptive & Explainable Data Fabrics. <https://lnkd.in/dVeegA4H>
- **Newsletter:** The Engineering Log by Akshay - ongoing LinkedIn publication on distributed systems, data engineering, and GenAI. [Read on LinkedIn](#)

CERTIFICATIONS

AWS	Solutions Architect Professional Data Analytics Specialty Security Specialty
Databricks	Databricks Certified Professional Spark Certified Associate Developer
Confluent (Kafka)	Confluent Certified Developer for Apache Kafka
Microsoft Azure	Microsoft Certified: Azure Fundamentals
DAMA	Certified Data Management Professional (CDMP)
IIT Roorkee	Generative AI Engineering Program (completed)
Anthropic	Claude 101 - Certificate of Completion

TECHNICAL SKILLS

Languages	Python, Java, Scala, SQL, PL/SQL, JavaScript, HTML
Backend & APIs	FastAPI, Pydantic, asyncio/httpx, SQLAlchemy, pytest, Flask, Spring Boot, Spring Cloud, Spring Security, Hibernate ORM, REST/OpenAPI, OAuth2, JWT, KeyCloak, EJB, JMS, React, Jenkins, Maven, Git, SonarQube
Databases & Search	Oracle, PostgreSQL, SQL Server, Elasticsearch/OpenSearch (index design, mappings, Query DSL, aggregations, ILM), Redis, Delta Lake, Vector Search (Azure AI Search)
Big Data & Analytics	Kafka, PySpark (Structured Streaming, Pandas UDFs), Pandas, Databricks, Delta Lake, Unity Catalog, Hive, HDFS, Hadoop, Medallion Lakehouse, Azure Synapse Analytics, Azure Data Factory, Microsoft Fabric (OneLake), AWS Glue, Data Governance, Microsoft Purview, S3, Azure Storage, MDM / Entity Resolution
Cloud & Containers	AWS, Azure, Kubernetes (AKS/EKS), Docker, Docker Compose, Helm, Databricks Asset Bundles, GitHub Actions, Azure Logic Apps, Azure Functions, AWS Lambda, API Gateway, Microsoft Sentinel (SIEM/SOAR), Log Analytics/KQL, Grafana
Generative AI	LLMs (GPT-4o, Claude), Agentic AI, Multi-Agent Orchestration (Agno), RAG, Embeddings & Vector Search, Spring AI, Agno, AI Guardrails, AWS Bedrock (AgentCore), Microsoft Foundry, Azure AI Search, Hugging Face Transformers, LLM Fine-tuning, Claude Code, OpenAI/Anthropic Python SDKs

PROFESSIONAL EXPERIENCE

Principal Architect — Unisys Corporation

May 2021 – Present

Architect (2021 – 2024); Principal Architect (2024 – Present)

Unisys CloudForte - Enterprise Monitoring & Security Product (Unisys Stealth Family)

Observability product in the CSPM/cybersecurity portfolio - monitors enterprise infrastructure and applications in real time, providing the telemetry foundation for anomaly detection, security monitoring, and threat visibility.

URL: [marketplace.microsoft.com — CloudForte for Azure](https://marketplace.microsoft.com/en-us/products/unisys-cloudforte)

- Designed the overall product architecture, cloud platform setup, and technology choices for CloudForte. Built the event-processing layer as microservices on Azure Kubernetes Service (AKS) - consuming ~5M events/day safely (no duplicates, automatic retries, dead-letter queues for failed messages) and triggering alerts or automated fixes based on risk. Designed multi-tenant isolation across the platform using per-tenant data partitioning in storage layer.
- Integrated CloudForte with Microsoft Sentinel — forwarding normalized security telemetry via Log Analytics ingestion and mapping platform alerts to Sentinel analytics rules and incidents — giving SOC teams unified threat visibility across infrastructure and application events.
- Designed the Elasticsearch/OpenSearch telemetry store — index templates and mappings, time-based indices with ILM rollover, and Query DSL aggregations for search and alerting — tuning shard sizing and query performance for high-volume event data.
- Designed an event-driven GenAI incident-triage agent - Python microservices consume Sentinel incidents and invoke an Azure AI Foundry agent grounded on runbook and incident-history knowledge sources — classifying incidents, recommending fixes, and estimating impact, with remediation executing only through approved, tested runbooks and results written back to the incident. Achieved 30% MTTR reduction.

Unisys Intelligence Accelerator (UIA) - Agentic AI Platform

Enterprise multi-agent platform automating IT operations, application development, and data tasks for large enterprises.

- Designed and built the agent runtime as FastAPI microservices exposing agent capabilities via REST; containerized with Docker and deployed on Kubernetes with GitHub Actions CI/CD and pytest-based test suites.
- Built a reusable Python agent harness on Amazon Bedrock AgentCore - agent lifecycle, tool/MCP registration, guardrails, and observability — with parity deployment on Microsoft Foundry; agents grounded via managed Knowledge Bases (RAG) using embeddings and vector search over enterprise documentation.

Banking Data Platform Modernization - Financial Services Client

- Designed the migration strategy after Microsoft deprecated the Export to Data Lake service - built a data-ingestion solution pulling Dynamics 365 F&O data from Microsoft Fabric (Fabric Link/OneLake) into an Azure Synapse dedicated SQL pool, with zero disruption to downstream financial reporting across ~100+ pipelines.

- Developed a PySpark-based solution in Azure Databricks (Delta Lake) to process Dynamics 365 F&O financial data, migrating ledger and financial-transaction processing into scalable, version-controlled Spark workloads - cutting pipeline runtime from ~3 hrs to 20 mins. Implemented Databricks Asset Bundles (DABs) with GitHub Actions CI/CD to declaratively define, promote, and deploy workloads across dev, staging, and prod environments. Implemented Databricks Lakehouse Federation to query external SQL sources in place through Unity Catalog, avoiding redundant ingestion pipelines.

Architect — Cognizant (Client: Novartis)

July 2018 – May 2021

Adverse Event resolution product for Novartis — analyses adverse events and predicts resolutions.

URL: <https://ae-service.novartis.net/pae-coreui>

- Architected and delivered a full-stack, event-driven system — extracting adverse-event data from Salesforce Social Studio streams and partner APIs, then cleaning, normalizing, deduplicating, and enriching it through microservices, AWS Lambda, API Gateway, and SQS over REST — driving cloud adoption that saved nearly \$1M annually.
- Led the Strategic Data Platform build-out, moving the stack from synchronous to asynchronous processing (SQS + Lambda) and designing an S3 data lake consolidating RDBMS and file-system sources with schema normalization and entity matching — solving critical data-latency problems for the data-ingestion product while driving Data Security, HIPAA, and GDPR compliance. Contributed hands-on across the full stack — UI, APIs, backend, and data pipelines.

Technical Manager — Capgemini (Client: Morgan Stanley)

Feb 2017 – June 2018

- Owned end-to-end product development - architecture, design, development, release, and maintenance of Spring Boot microservices: defined service boundaries and API contracts, implemented REST controllers, service and DAO layers, and designed entities and the underlying data model, and full stack development using React JS, Spring boot.
- Designed the microservices architecture with clear separation of concerns — stateless services, layered design (controller/service/repository), centralized configuration, and inter-service communication over REST - enabling independent development, testing, and release of services.

Technical Lead — Persistent Systems (Clients: Thermo Fisher, Proofpoint)

Sept 2014 – Jan 2017

Thermo Fisher — Ion Reporter: NGS product automating sequencing, analysis, and reporting. URL:

<https://ionreporter.thermofisher.com>

- Led backend development of REST APIs and Spring/Hibernate persistence layers for the sequencing analysis workflow; designed batch processing for large result sets, cutting report-generation time.

Proofpoint — Enterprise email security and compliance platform (threat protection, DLP, archiving).

- Built high-volume message-processing and policy-management services (Spring, Hibernate, REST) for threat-detection and DLP/archiving modules, with idempotent processing, RBAC, and audit trails for regulated customers.

Earlier Roles — Worldline (BNP Paribas), Polaris (JPMorgan), HAL

Jan 2009 – Aug 2014

- Built Spring/Hibernate transaction engines for SEPA direct-debit and credit-transfer processing (Worldline); designed EJB/JMS multi-tier enterprise platforms for JPMorgan's TLIP liquidity application (Polaris); developed an in-house inventory and supply-chain ERP with batch processing (HAL).

KEY ENGINEERING HIGHLIGHTS

Microservices & Backend

- Backend Stack Ownership: Took complete ownership of the backend stack - from development through cloud deployment - for 30+ microservices serving enterprise applications across multiple organizations; ensured resiliency via circuit breakers, bounded retries with exponential backoff, dead-letter queues, zero-downtime deployments on Kubernetes (AKS/EKS).
- Event-Driven Consistency: Built SQS-based microservices using the transactional outbox pattern - saving business data and the outgoing event in one database transaction before publishing
- Python Async Services: designed FastAPI services with asyncio/httpx for non-blocking I/O, Pydantic request/response validation, OpenAPI-driven contracts, and Kafka/SQS consumers in Python for asynchronous, distributed processing.

Distributed Systems Design

- High-Throughput Partitioning: diagnosed and fixed hot-partition skew in high-velocity Kafka pipelines by redesigning topic partition keys (entity-level composite keys).
- Fault-Tolerant Design: designed cloud data-ingestion services for partition tolerance - idempotent, retry-safe operations with exponential backoff and jitter, plus dead-letter queues.

Massive-Scale Data Ingestion, Engineering and Governance

- Petabyte-Scale Ingestion: architected a high-velocity ingestion engine on Spark handling petabytes of records.

- Low-Latency Streaming: implemented event-driven Kafka pipelines with fault-tolerant consumer-group services (Java and Python) — cutting data availability to minutes. Implemented entity resolution and golden-record creation (MDM) with Azure Purview for governance and lineage.

DevOps & CI/CD

- Infrastructure as Code: architected Databricks Asset Bundles to declaratively define, version, and promote data workloads across environments - integrated with secure GitHub Actions pipelines on VNet-isolated self-hosted runners with strict network boundaries, environment-specific variable injection. Used ADF arm template for Pipeline deployment.
- Containerized Orchestration: containerized distributed applications with Docker and orchestrated fault-tolerant deployments on managed Kubernetes (AKS), leveraging auto-scaling, rolling updates, and self-healing.

Generative AI

- Designed and developed agents on Azure AI Foundry and Amazon Bedrock AgentCore — coordinating specialized retrieval, reasoning, and execution agents — and applied LLMs, embeddings, and vector search to data problems: incident classification, entity matching, and automated data enrichment.

AWARDS & RECOGNITION

- **Unisys All-Hands Recognition:** recognized four times at company-wide All-Hands meetings for high-impact work across distributed systems and data engineering initiatives.
- **“You Rock” Award Nomination — Persistent:** nominated by peers and leadership for consistent technical excellence.

EDUCATION

- Bachelor of Engineering (B.E.), Computer Science — V.Y.W.S. College of Engineering, Badnera, Amravati University (1st Division).
- Generative AI Engineering Program — IIT Roorkee.